CSE221: Algorithms
Lab-08
Summer 2026

Announcements for Students

This is your last lab. There will be no lab class or lab quiz next week.

The Lab 06 Quiz marks will be published by this Monday. Also, once the remaining lab assignment deadline has passed, your total lab assignment marks will also be shown.

You can view your lab marks from the lab portal [click here]. If you find any discrepancies, please inform your lab faculties as soon as possible. No further concerns will be considered after 05:00 PM on September 06.

Please note that, in accordance with the lab policy, the total marks may still CHANGE for students who are found to have been involved in any unfair means. Our plagiarism checking process is still ongoing, so the marks of those students may be revised AT ANY TIME based on the findings.

Assignment-08 Links:

Codeforces Invitation Link:
https://codeforces.com/contestInvitation/a9e70299c5c814cb78ded8d592f2d67721bb43c9

Alternative Link: https://codeforces.com/contests/713166
[The alternative link will **only work** after you have successfully registered using the first link.]

Deadline: Sep 6, 2026 2:00 PM

Note: Use FAST I/O in your code to avoid Time Limit Exceeded (TLE), especially in the quiz.

After the deadline ends, you can submit your solution by entering the contest as a Virtual Participant for practice purposes. **However, please note, submissions after the deadline are not accepted for grading.**

Please note that your assignment submissions will not be evaluated if they are flagged as AI-generated.

Learning Outcome:

By the end of this lab, students will be able to identify, and solve optimization problems using greedy algorithm techniques, particularly in graph-based and scheduling scenarios. They will understand the greedy-choice property and how locally optimal decisions can lead to globally optimal solutions in problems such as Minimum Spanning Tree (MST) construction and interval task scheduling. Students will also develop an understanding of Disjoint Set Union (DSU) as the underlying algorithmic mechanism used to maintain connectivity during MST construction.
Overall, the lab aims to strengthen students' ability to recognize greedy problem structures, reason about correctness, and design efficient solutions for real-world optimization problems.